The examination is being carried out on the **following application documents**

Description, Pages

1-11 as published

Claims, Numbers

1-19 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/6-6/6 as published

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 Vivek Dham: "Programming Chirp Parameters in TI Radar Devices", , 1 May 2017 (2017-05-01), pages 1-16, XP093159247, TI web site Retrieved from the Internet: URL:https://www.ti.com/lit/an/swra553a/swra553a.pdf?ts=1714779247240&ref_url=https%253A%252F%252Fwww.google.com%252F

D2 CN 101 089 653 B (UNIV XI AN TECHNOLOGY) 9 March 2011 (2011-03-09)

The following document has been cited in the international search report; the numbering will be adhered to in the rest of the procedure.

D3 US 8 731 502 B2 (FREESCALE SEMICONDUCTOR, INC.) 20 May 2014 (2014-05-20)

1 Unity (Art. 82 EPC)

1.1 The application lacks unity within the meaning of Article 82 EPC.

1.2 This Authority considers that the application does not meet the requirements of unity of invention and that there are 3 inventions covered by the claims indicated as follows (in bold the distinctive special technical features of each inventions):

1.2.1 Group I: claims 1, 2, 5, 7, 8-12, 14, 17-19

"*A frequency modulated continuous wave (FMCW) synthesizer, comprising:*
a phase locked loop (PLL) including:
a frequency divider including a control input, a frequency divider input, and a frequency divider output, and configured to modify a frequency of a signal, received by the frequency divider input, in response to the control input;
a control voltage generator (CVG) including a first CVG input, a second CVG input, and a control voltage output, the first CVG input adapted to receive a frequency reference signal, the second CVG input coupled to the frequency divider output, the CVG configured to generate a control voltage in response to the first CVG input and the second CVG input; and
a voltage controlled oscillator (VCO) including a VCO input and a VCO output, the VCO input coupled to the control voltage output, the VCO output coupled to the frequency divider input, the VCO configured to output a FMCW output signal having a FMCW output frequency in response to the VCO input; and
a control engine including a control output coupled to the control input of the frequency divider, the control engine configured to control the PLL so that the FMCW output frequency:
from a first time to a second time, is a first idle frequency;
from the second time to a third time, changes at a specified first rate;
from the third time to a fourth time, changes at a specified second rate, wherein the first rate is different from the second rate; and
from the fourth time to a fifth time, is a second idle frequency.

further comprising:
a receiver adapted to receive a reflected FMCW chirp, the receiver including a receiver output;
an analog to digital converter (ADC) including an ADC input and an ADC output, the ADC input coupled to the receiver output, the ADC configured to sample signals received by the ADC input and to output the sampled signals;
***a processor** including a processor input coupled to the ADC output, the processor configured to, in response to the sampled signals sampled between the second time and the third time, determine at least one of presence, distance, or velocity of target objects, and **configured not to perform the determine action between the third time and the fourth time;** and*
a transmitter including a transmitter input and configured to transmit signals received by the transmitter input, the transmitter input coupled to the VCO output."

1.2.2 Group II: claims 3, 4, 6, 13, 15

"*A frequency modulated continuous wave (FMCW) synthesizer, comprising:*
a phase locked loop (PLL) including:
a frequency divider including a control input, a frequency divider input, and a frequency divider output, and configured to modify a frequency of a signal, received by the frequency divider input, in response to the control input;
a control voltage generator (CVG) including a first CVG input, a second CVG input, and a control voltage output, the first CVG input adapted to receive a frequency reference signal, the second CVG input coupled to the frequency divider output, the CVG configured to generate a control voltage in response to the first CVG input and the second CVG input; and
a voltage controlled oscillator (VCO) including a VCO input and a VCO output, the VCO input coupled to the control voltage output, the VCO output coupled to the frequency divider input, the VCO configured to output a FMCW output signal having a FMCW output frequency in response to the VCO input; and
a control engine including a control output coupled to the control input of the frequency divider, the control engine configured to control the PLL so that the FMCW output frequency:
from a first time to a second time, is a first idle frequency;
from the second time to a third time, changes at a specified first rate;
from the third time to a fourth time, changes at a specified second rate, wherein the first rate is different from the second rate; and
from the fourth time to a fifth time, is a second idle frequency.

*wherein the CVG includes a a filter that includes a filter signal input**, a filter control input**, and a filter output:*"

It is noted that implementing "***a filter control input"***, as claimed in claim 3 **,** is directly related (and considered aspects of the same special technical feature) to the control engine controlling the CVG current in claim 6 and filter bandwidth variations claimed in claim 13 and 15 that are consequently considered in the reported group of invention II.

1.2.3 Group III: claim 16

"*A method for operating a frequency modulated continuous wave (FMCW) radar, comprising:*
receiving, by a receiver, a FMCW chirp reflection signal;
generating, using a FMCW synthesizer, a FMCW synthesizer signal at a FMCW synthesizer output frequency, wherein the FMCW synthesizer output frequency:
from a first time to a second time, is a first FMCW idle frequency;
from the second time to a third time, changes at a specified first rate;
from the third time to a fourth time, changes at a specified second rate, wherein

the first rate is different from the second rate; and
from the fourth time to a fifth time, is a second FMCW idle frequency;
mixing, using a mixer, the FMCW chirp reflection signal and the FMCW synthesizer signal, to generate a mixer output;
sampling the mixer output using an analog-to-digital converter (ADC); and
transmitting, using a transmitter, the FMCW synthesizer signal.

further comprising:
lowering a current of an output of a charge pump of a phase locked loop (PLL) of the FMCW synthesizer to be relatively lower between the second time and the third time; and
raising a current of the output of the charge pump of the PLL of the FMCW synthesizer to be relatively higher between the third time and the fourth time."

1.2.4 The single general concept of the three groups is:

"A frequency modulated continuous wave (FMCW) synthesizer, comprising:
a phase locked loop (PLL) including:
a frequency divider including a control input, a frequency divider input, and a frequency divider output, and configured to modify a frequency of a signal, received by the frequency divider input, in response to the control input;
a control voltage generator (CVG) including a first CVG input, a second CVG input, and a control voltage output, the first CVG input adapted to receive a frequency reference signal, the second CVG input coupled to the frequency divider output, the CVG configured to generate a control voltage in response to the first CVG input and the second CVG input; and
a voltage controlled oscillator (VCO) including a VCO input and a VCO output, the VCO input coupled to the control voltage output, the VCO output coupled to the frequency divider input, the VCO configured to output a FMCW output signal having a FMCW output frequency in response to the VCO input; and
a control engine including a control output coupled to the control input of the frequency divider, the control engine configured to control the PLL so that the FMCW output frequency:
from a first time to a second time, is a first idle frequency;
from the second time to a third time, changes at a specified first rate;
from the third time to a fourth time, changes at a specified second rate, wherein the first rate is different from the second rate; and
from the fourth time to a fifth time, is a second idle frequency."

1.2.5 This concept is known in the art, for example in the prior art identified as document D3 that discloses:

"A frequency modulated continuous wave synthesizer (claim 2 "generate a low-path control signal for providing Frequency Modulated Continuous Wave (FM-CW) modulation")*, comprising:*
a phase locked loop (fig. 1) *including:*
a frequency divider (fig. 1 ref. 120) *including a control input, a frequency divider input* (fig. 1 ref. 137)*, and a frequency divider output, and configured to modify a frequency of a signal, received by the frequency divider input, in response to the control input;*
a control voltage generator (fig. 1 ref. "Phase detector" and filter with ref. 117 output) *including a first CVG input, a second CVG input* (see fig. 1 ref. "Phase detector" inputs)*, and a control voltage output* (fig. 1 ref. 117)*, the first CVG input adapted to receive a frequency reference signal* (fig. 1 ref. Fref)*, the second CVG input coupled to the frequency divider output* (fig. 1 ref. 120 output)*, the CVG configured to generate a control voltage in response to the first CVG input and the second CVG input* (i.e obvious functionality of disclosed CVG)*; and*
a voltage controlled oscillator including a VCO input and a VCO output (fig. 1 ref. 110)*, the VCO input coupled to the control voltage output, the VCO output coupled to the frequency divider input* (see fig. 1 ref. 110 output to ref. 120 input)*, the VCO configured to output a FMCW output signal having a FMCW output frequency in response to the VCO input* (fig. 3 shows the FMCW output of the disclosed synthetizer)*; and*
a control engine (fig. 1 ref. 125, 130; fig. 8 ref. 825, 830) *including a control output* fig. 1 ref. 137) *coupled to the control input of the frequency divider, the control engine configured to control the PLL so that the FMCW output frequency:*

from a first time (fig. 3 ref. 310) *to a second time* (fig. 3 ref. 330)*, is a first idle frequency* (fig. 3 ref. F_1)*;*
from the second time (fig. 3 ref. 330) *to a third time* (fig. 3 ref. 340)*, changes at a specified first rate* (see fig. 3 rising slope)*;*
from the third time to a fourth time (fig. 3 ref. 360)*, changes at a specified second rate* (see fig. 3 falling slope)*, wherein the first rate is different from the second rate* (col. 5 lines 50-56 "When the frequency modulation pattern 400 Subsequently reaches the new high frequency limit value F (re-programmed at point 440) at point 470, the frequency modulation pattern transitions to a falling slope. **The falling slope now comprises a slope gradient defined by the (once again) re-programmed ramp slope value RS, which for the illustrated example is steeper than the previous gradient**")*; and*
from the fourth time (fig. 3 ref. 360) *to a fifth time, is a second idle frequency* (fig. 3 ref. F_1)*."*

1.2.6 It is noted that the subject-matter of claims 10, 14, the independent claims on which claims 13 and 16 are respectively dependent, is equivalent to claim 1 representing the single general concept discussed above. Hence, the reported features analysis of D3 related to claim 1 formulation of teh single general concept is valid, mutatis mutandis, for claim 10 and 14 as well. Also, the mixer claimed only independent claim 14 (and not in independent claims 1, 10) is also disclosed in D3 (see D3 fig, 10 ref. 1030).

1.2.7 Hence, the three groups of inventions are not so linked as to form a single general inventive concept.

1.3 Furthermore, the technical effects of the three inventions and the problems solved by the three inventions cannot be considered as to be corresponding, for the following reason:

1.4 Invention I is directed to a frequency modulated continuous wave synthesizer with improved power consumption (i.e. by not performing determined processing in time window where the received signal is not relevant).

1.5 Invention II is directed to a frequency modulated continuous wave synthesizer with wider frequency range (i.e. by having a dedicated control signal for the PLL filter).

1.6 Invention III is directed to a frequency modulated continuous wave synthesizer with faster settling (i.e. by having accordingly programming of the charge pump deliverable current).

1.7 The subject-matter to be excised may be made the subject of one or more divisional applications according to Rule 36 EPC.

2 Novelty of independent claims (Art. 52(1) and 54 EPC)

Independent claims 1, 10, 14

2.1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 is not new within the meaning of Article 54(1) and (2) EPC.

2.2 Document D3 discloses the subject-matter of claim 1 as reported in 1.2.5 above.

2.2.1 The subject-matter of claim 1 is therefore not new (Article 54(1) and (2) EPC) versus D3.

2.3 Claim 1 is also not new, in the meaning of Article 54 EPC, versus document D2 that discloses a Frequency Modulated Continuous Wave (FMCW) mmWave radar (D2 claim 1 " frequency-modulation continuous wave FMCW radar ") where, as standard in the art, a PLL (D2 fig. 5, 6) is implemented to generate

chirps signals and where said chirps signals are built on four different times with the distribution of idle frequencies phases and frequency ramp phases with different rates as claimed in claim 1 (D2 fig. 3 explicitly discloses said chirps settings). Furthermore, the PLL disclosed in D2 is explicitly disclosed with all the typical features comprised in standard PLL and claimed in claim 1 (see D2 par. 0071 "thereby taking out its difference frequency, dividing it in the phase-locked loop unit (PLL) and performing phase detection with the DDS reference signal. The output signal controls the output frequency of the voltage-controlled oscillator after passing through the loop filter, so that it changes according to the set frequency")

2.4 Claim 1 is also not inventive, in the meaning of Article 56 EPC, versus document D1 that discloses a Frequency Modulated Continuous Wave (FMCW) mmWave radar (D1 page 2 par. 1 "Frequency Modulated Continuous Wave (FMCW) mmWave radar") where, as standard in the art, a PLL (D1 page 4 par. 2.2.1 "Also the closed loop PLL is designed to support a very fast settling of the frequency ramp.") is implemented to generate chirps signals and where said chirps signals are built on four different times with the distribution of idle frequencies phases and frequency ramp phases with different rates as claimed in claim 1 (D1 fig. 1 and 2 explicitly discloses said chirps settings).

2.4.1 D1 is silent on the implementation of the PLL where the claimed *control voltage generator* (corresponding to, e.g, standard PLL charge pumps and loop filter), *VCOs, control engines* (corresponding to, e.g., frequency divider control logic in every PLL frequency synthesizer) and related claimed connections (whose technical effect is allowing the implementation of a working phase locked loop) is considered the most straightforward solution in the art that the person skilled in the art would first consider when facing the task of implementing an actual PLL starting from D1 without involving an inventive step.

2.5 Following, mutatis mutandis, the reasoning above indepedent claims 10, 14 are also not novel versus documents D2, D3 and not inventive versus D1. It is noted that the mixer claimed only independent claim 14 (and not in independent claims 1, 10) is also disclosed in D2 (D2 fig. 7 and par. 0072 "Figure 7, which is mainly used to process the echo signal, filter out the interference of other radars, and obtain the high-resolution distance and speed of the target. The echo and interference are low-noiseamplified and mixed with the local oscillator signal in the **mixer**") and D3 (see D3 fig. 10 ref. 1030).

2.6 **Dependent claims**

Dependent claims 2, 5, 7-9, 11, 12, 17-19

Dependent claims *2, 5, 7-9, 11, 12, 15, 17-19* do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step.

Claims 2, 11 -> anticipated by D1 that discloses:

+ a receiver adapted to receive a reflected FMCW chirp, (D1 figure 3 and page 5 line 7 "the spacing of the receiver antenna")
+ an analog to digital converter (ADC) (D1, e.g., page 3 line 17 "ADC sampling frequency").

+ the processing related to ADC for determining presence, distance, or velocity of target objects action not performed in the third time and the fourth time (see fig. 1 where the "ADC start" is set after the correspondent fourth time, i.e. at the end of the explicitly disclosed "idle time");

+ standard transmitter as claimed in claim 2 (D1, e.g., page 11 four lines from the bottom "The different chirps can also use different transmitters").

+ as claim 11 depends on claim 2 it is noticed that the second rate being greater the the first rate is also disclosed in D1 fig. 1 and 2. Hence, claim 11 is anticipated by claim 11.

Claim 5 -> explicitly anticipated by D2 (D2 fig. 3,4 constant frequency slopes), D3 (D3 fig. 4 constant frequency slopes).

Claims 7, 17 -> explicitly anticipated by D2 (D2 fig. 3 different idles frequencies) and considered non inventive versus D1 (that also discloses a parameter for setting the idle time, see Table 2) and D3 as being a possible alternative not requiring an inventive step to the person skilled in the art starting from D1 or D3.

It is also noticed that the claimed third and fourth rate are explicitly disclosed in D1 fig. 1 and 2, D2 fig. 3, 4 and D3 fig. 4.

Claims 8, 19 -> the options of the time steps sequences as claimed in claims 8 and 19 is fully correspondent to the sequences disclosed in the reported figures of the three documents D1 (D1 fig. 1, 2), D2 (D2 fig. 3), D3 (D3 fig. 4).

Claim 9 -> explicitly anticipated by D3 (D3 fig. 8 feedback from ref. 820 to "control engine" ref. 830) and considered non inventive versus D1 and D2 as being a possible alternative not requiring an inventive step to the person skilled in the art starting from D1 or D2.

Claims 12, 18 -> anticipated by each of D1, D2 or D3 as the claimed subject-matter is an intrinsic functionality of the disclosed PLLs.